

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

PACIFIC BIOSCIENCES OF CALIFORNIA, INC.,
Petitioner,

v.

PERSONAL GENOMICS TAIWAN, INC.,
Patent Owner.

IPR2020-01200
Patent 7,767,441 B2

Before ERICA A. FRANKLIN, SUSAN L. C. MITCHELL, and MICHAEL
VALEK, *Administrative Patent Judges*.

MITCHELL, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. *Background and Summary*

On June 27, 2020, Pacific Biosciences of California, Inc. (“Petitioner”) filed a Petition (Paper 1, “Pet.”) requesting *inter partes* review of claims 1, 2, 6, 7, 10–22, 24, and 27–36 (“the challenged claims”) of U.S. Patent No. 7,767,441 B2 (Ex. 1001, “the ’441 patent”). See 35 U.S.C. §§ 311–319. On October 22, 2020, Personal Genomics Taiwan, Inc. (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 13 (“Prelim. Resp.”). On January 19, 2021, we granted institution of an *inter partes* review. See Paper 15, 3–4, 35 (“Dec.”). Patent Owner filed a Response on April 13, 2021. Paper 19 (“Resp.”). Petitioner filed a Reply on July 6, 2021. Paper 21 (“Reply”). Patent Owner filed its Sur-Reply on August 17, 2021. Paper 22 (“Sur-Reply”). An oral hearing was held on October 21, 2021. Paper 30 (“Tr.”).

This is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the challenged claims on which we instituted trial. Based on the complete record before us, we determine that Petitioner has not shown, by a preponderance of the evidence, that any of claims 1, 2, 6, 7, 10–22, 24, or 27–36 are unpatentable.

B. *Real Parties in Interest*

Each party identifies itself as the real party-in-interest. Pet. 5; Paper 4, 2.

C. *Related Proceedings*

The parties identify an ongoing court proceeding involving the ’441 patent: *Personal Genomics Taiwan, Inc. v. Pacific Biosciences of*

California, Inc., No. 19-cv-1810-LPS (D. Del. filed Sept. 26, 2019). Pet. 5, Paper 4, 2.

On June 22, 2020, Petitioner filed a first petition for *inter partes* review of claims 1–6, 9, and 43–58 of the ’441 patent. *See Pacific Biosciences of California, Inc. v. Personal Genomics Taiwan, Inc.*, IPR2020-01163, Paper 1 (PTAB June 22, 2020) (“1163 Petition”). We granted institution of an *inter partes* review as to claims 1–6, 9, and 43–58. *See* IPR2020-01163, Paper 15, 3–4, 38 (PTAB January 19, 2021).

D. The ’441 Patent

The ’441 patent, titled “Bioassay System Including Optical Detection Apparatuses, and Method for Detecting Biomolecules,” generally relates to a bioassay system including a plurality of optical detection apparatuses that can sense the existence of a fluorophore on a single molecule. *See* Ex. 1001, code (54), 3:55–61. The ’441 patent describes the need for a device that should be “capable of sequencing single molecules to avoid the known difficulty of asynchrony in both the amplification (e.g., drift between the sequences of ideally clonal templates) and sequencing (e.g., dephasing of the stepwise sequencing reactions amongst the sequencing templates) steps of clustered sequencing methods.” *Id.* at 1:67–2:6. The ’441 patent states that “[t]he bioassay system provided by the invention is capable of large-scale parallel sequencing reactions, i.e., simultaneously sequencing a large number of different nucleic acid templates. Each sequencing reaction uses a single molecule as the template (i.e., single molecule sequencing).” *Id.* at 2:13–18. The invention provides “direct correspondence of linker sites to which the nucleic acids being detected are attached . . . and one or more

detecting units (e.g., light detectors), and in part, on the short distance between the linker sites and the detecting units.” *Id.* at 2:21–27.

Figure 2 of the ’441 patent is reproduced below.

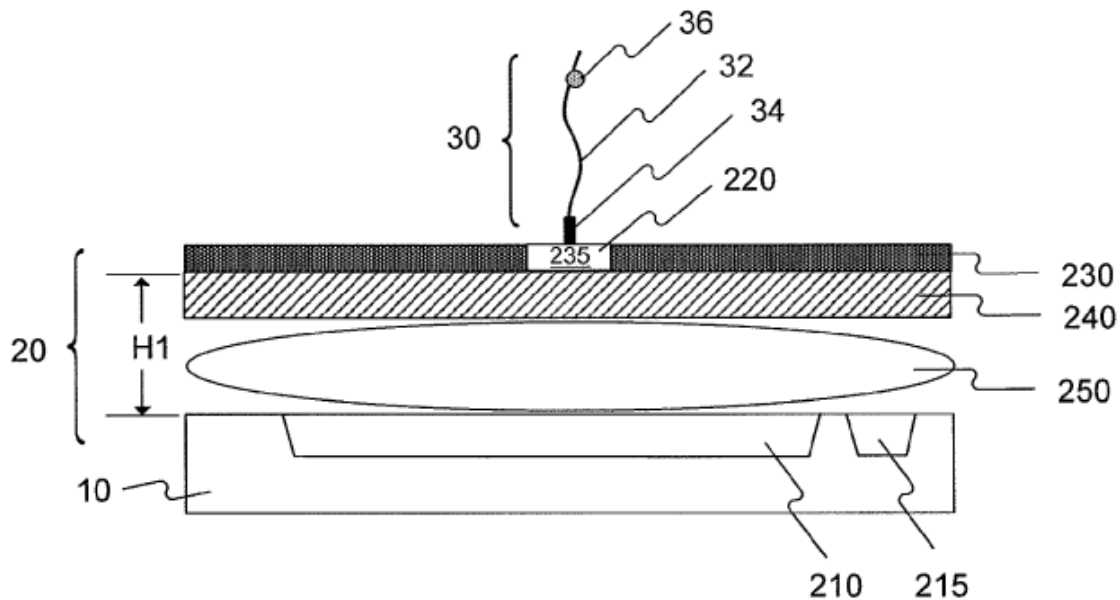


Fig. 2

Figure 2 above illustrates an optical detection apparatus. Ex. 1001, 3:24–26. Optical detection apparatus 20 includes light detector 210 formed on substrate 10. *Id.* at 4:52–60. Light detector 210 may alternatively include one or more photoconductive photon detectors, photovoltaic photon detectors, or photodiodes. *Id.* at 5:1–11. Optical detection apparatus 20 may also include blind sheet 230 over light detector 210, including pinhole 235 having a diameter of less than 1,000 nanometers. *Id.* at 5:12–22. The ’441 patent further states that “linker site 220 formed proximate to pinhole 235 may be spaced apart from light detector 210 by a distance H1 of less than or equal to 100 micrometers.” *Id.* at 5:25–28. Optical detection apparatus 20 may optionally include filter layer 240 and microlens 250 between light

detector 210 and blind sheet 230. *Id.* at 5:31–33. Filter layer 240 may include one or more transparent sublayers. *Id.* at 5:36–38.

The '441 patent states that “linker site 220 may be treated to affix a single biomolecule 30 thereto.” Ex. 1001, 5:49–50. For example, biomolecule 30 may include single stranded DNA molecule 32, affixed to linker site 220 with end link primer 34. *Id.* at 5:50–54. DNA molecule 32 may be labeled with fluorophore 36. *Id.* at 5:54–55. The '441 patent also states that “[w]hen excited by excitation light of a first wavelength λ_1 , fluorophore 36 may emit fluorescent light of a second wavelength λ_2 .” *Id.* at 5:55–57. Furthermore, “[l]ight detector 210 then detects the fluorescent light emitted from fluorophore 36, so as to identify the type of base that fluorophore 36 is attached to, thereby sequentially determining the sequence of DNA molecule 32.” *Id.* at 5:60–64.

E. Challenged Claims

Petitioner challenges claims 1, 2, 6, 7, 10–22, 24, and 27–36 of the '441 patent. Claims 1, 2, and 6 are also challenged in IPR2020-01163.

Claims 1 and 12 are independent. Claims 2, 6, 7, and 10 depend from claim 1. Claim 10 recites an optical detection system comprising at least 10,000 apparatuses as recited by claim 1. Claims 16 and 30 recite methods for sequencing a nucleic acid or detecting a biomolecule, respectively, using the apparatus of claim 1.

Claim 1 is illustrative and recites:

1. An apparatus for identifying a single biomolecule, comprising:
 - a substrate having a light detector; and
 - a linker site formed over the light detector, the linker site being treated to affix the biomolecule to the linker site;

wherein the linker site is proximate to the light detector and is spaced apart from the light detector by a distance of less than or equal to 100 micrometers.

Ex. 1001, 26:11–19.

F. Prior Art and Asserted Grounds of Unpatentability

Petitioner asserts that claims 1, 2, 6, 7, 10–22, 24, and 27–36 would have been unpatentable based on the following grounds:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36	102(b) ¹	Hassibi ²
1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36	103(a)	Hassibi in view of the skill of a POSA (Person of Ordinary Skill in the Art)
17, 20	103(a)	Hassibi and Korlach ³
28, 29	103(a)	Hassibi in view of the skill of a POSA
21, 22, 24	103(a)	Hassibi and Cheeseman ⁴
1, 2, 6	102(b)	Blumenfeld ⁵
1, 2, 6	103(a)	Blumenfeld in view of the skill of a POSA

¹ The Leahy-Smith America Invents Act (“AIA”) included amendments to 35 U.S.C. §§ 102 and 103. Because the ’441 patent issued from an application filed before the effective date of these amendments, we apply the pre-AIA versions of §§ 102 and 103.

² Hassibi et al., US 2004/0197793 A1, published Oct. 7, 2004 (Ex. 1005, “Hassibi”).

³ Korlach et al., US 7,361,466 B2, issued April 22, 2008 (Ex. 1006, “Korlach”).

⁴ Cheeseman, US 5,302,509, issued April 12, 1994 (Ex. 1040, “Cheeseman”).

⁵ Blumenfeld et al., US 2002/0018199 A1, published Feb. 14, 2002 (Ex. 1007, “Blumenfeld”).

Petitioner submits the Declaration of Dean P. Neikirk, Ph.D. in support of its positions. *See* Ex. 1003. Patent Owner submits the Declaration of Timothy D. Harris, Ph.D. *See* Ex. 2008.

II. ANALYSIS

A. *Principles of Law*

1. *Burden*

In an *inter partes* review, the burden of proof is on the Petitioner to show that the challenged claims are unpatentable, and that burden never shifts to the patentee. *See* 35 U.S.C. § 316(e); *In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1375 (Fed. Cir. 2016) (citing *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015)).

2. *Anticipation*

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). “A reference anticipates a claim if it discloses the claimed invention ‘such that a skilled artisan could take its teachings in combination with his own knowledge of the particular art and be in possession of the invention.’” *In re Graves*, 69 F.3d 1147, 1152 (Fed. Cir. 1995) (internal citation and emphasis omitted). Moreover, “it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968); *see Eli Lilly and Co. v. Los Angeles Biomedical Res. Inst. at Harbor-UCLA Medical Center*, 849 F.3d 1073, 1074–75 (Fed. Cir. 2017).

An apparatus claim covers “what a device *is*, not what a device *does*.” *ParkerVision, Inc v. Qualcomm Inc.*, 903 F.3d 1354, 1361 (Fed. Cir. 2018) (citation omitted).

As a result, “[a]n invention need not *operate* differently than the prior art to be patentable, but need only *be* different”—or, rather, “*unobviously* different.” A corollary of these principles is that an apparatus that is “capable of” performing certain functions may be anticipated by or obvious in view of a prior art apparatus that can likewise perform these functions. Indeed, “depending on the claims, ‘an accused device may be found to infringe if it is reasonably capable of satisfying the claim limitations, even though it may also be capable of noninfringing modes of operation. Similarly, a prior art reference may also anticipate or render obvious an apparatus claim—depending on the language of the claim—if the reference discloses an apparatus that is reasonably capable of operating so as to meet the claim limitations, even if it does not meet the claim limitations in all modes of operation.

Id. (citations omitted).

3. *Obviousness*

A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains (“POSA” or “POSITA”). *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4)

objective evidence of nonobviousness.⁶ *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). In determining obviousness when all elements of a claim are found in various pieces of prior art, “the factfinder must further consider the factual questions of whether a person of ordinary skill in the art would be motivated to combine those references, and whether in making that combination, a person of ordinary skill would have had a reasonable expectation of success.” *Dome Patent L.P. v. Lee*, 799 F.3d 1372, 1380 (Fed. Cir. 2015); *see also WMS Gaming, Inc. v. Int’l Game Tech.*, 184 F.3d 1339, 1355 (Fed. Cir. 1999) (“When an obviousness determination relies on the combination of two or more references, there must be some suggestion or motivation to combine the references.”). “Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant’s disclosure.” *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

An obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that

⁶ Patent Owner also does not present any objective evidence of nonobviousness. *See generally* Resp.

instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. “*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless ‘the improvement is more than the predictable use of prior art elements according to their established functions.’” *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with the above-stated principles.

B. Level of Ordinary Skill in the Art

We consider the asserted grounds of unpatentability in view of the understanding of a person of ordinary skill in the art. *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 399 (2007) (stating that obviousness is determined against the backdrop of the scope and content of the prior art, the differences between the prior art and the claims at issue, and the level of ordinary skill in the art). Factual indicators of the level of ordinary skill in the art include “the various prior art approaches employed, the types of problems encountered in the art, the rapidity with which innovations are made, the sophistication of the technology involved, and the educational background of those actively working in the field.” *Jacobson Bros., Inc. v. U.S.*, 512 F.2d 1065, 1071 (Ct. Cl. 1975); *see also Orthopedic Equip. Co. v. U.S.*, 702 F.2d 1005, 1011 (Fed. Cir. 1983) (quoting with approval *Jacobson Bros.*).

In our Decision on Institution, we applied Petitioner’s proposed definition of one skilled in the art as an individual with “(1) at least [a] Bachelor of Arts or Bachelor of Science degree in biology, biochemistry,

applied physics, or a similar degree,” and (2) “at least 1-2 years of experience in the field of optics and sensors.” Dec. 10.

In the Response, Patent Owner argues that the person of ordinary skill in the art should also have 1–2 years of experience in “molecular sequencing, including handling and sequencing of nucleic acids.” Resp. 2 (citing Ex. 2008 ¶ 23). Patent Owner supports this contention by citing to various discussions throughout the Specification of the ’441 patent referring to the sequencing of nucleic acids. *See id.* at 2–3 (citing Ex. 2008 ¶¶ 16–23). Patent Owner also contends that the language recited in the claims supports this definition. *Id.* at 3.

Patent Owner argues that Petitioner’s expert, Dr. Neikirk, does not have the corresponding experience in molecular sequencing. *See* Resp. 3. Accordingly, Patent Owner contends that Dr. Neikirk’s opinions relating to molecular sequencing in the claims and the prior art, “should be accorded little or no weight.” *Id.* at 3–4.

We find Petitioner has the better argument. The claims at issue in this *inter partes* review involve a molecular sensor, its components, or a method of using or manufacturing the sensor. *See* Ex. 1001, 26:11–37, 27:64–30:6. An in-depth understanding of molecular sequencing would not be required to understand these claims, the ’441 patent, the prior art asserted against these claims, or the problems solved by the claimed apparatus and methods. Therefore, we will continue to apply Petitioner’s definition of one of skill in the art as set forth above.

Moreover, even under Patent Owner’s proposal for the level of ordinary skill, we find that Dr. Neikirk is qualified to offer the opinions in his declaration. “There is . . . no requirement of a perfect match between the

expert’s experience and the relevant field” and one need “not be a person of ordinary skill in the art in order to testify as an expert under Rule 702” so long as they are “qualified in the pertinent art.” PTAB Consolidated Trial Practice Guide (Nov. 2019) 34 (citing precedent). Here, the record demonstrates that Dr. Neikirk is qualified in the art of molecular sensors. *See* Ex. 1003 ¶¶ 4–20; Ex. 1004 (describing Dr. Neikirk’s technical background and expert qualifications). Therefore even if one of ordinary skill would have had experience in molecular sequencing, as Patent Owner proposes, it does not follow that all of the testimony in Dr. Neikirk’s declarations should be accorded “little or no weight.”

C. Claim Construction

We construe claims using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 42.100 (2019). Therefore, we construe the challenged claims under the framework set forth in *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–19 (Fed. Cir. 2005) (en banc). Under this framework, claim terms are given their ordinary and customary meaning, as would be understood by a person of ordinary skill in the art, at the time of the invention, in light of the language of the claims, the specification, and the prosecution history of record. *Id.* Only those terms that are in controversy need be construed and only to the extent necessary to resolve the controversy. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Matal*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (citing *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

At the preliminary stage of the proceeding, Petitioner did not offer any express claim constructions for any claim term, *see* Pet. 10, but Patent

Owner did, *see* Prelim. Resp. 15–27. Patent Owner asserted that the preambles of the claims calling for a “identifying a single biomolecule” are limiting, and each of the claims requires a “linker site” to which a “single biomolecule” is affixed. Prelim. Resp. 15–27.

In our Institution Decision, we did not determine whether the preamble limits the claims to identifying a single biomolecule because we found that, on the preliminary record, Hassibi discloses an apparatus for identifying a single biomolecule. Dec. 12. We did interpret, however, the claims phrase “to affix *the* biomolecule” as referring to affixing “a single biomolecule.” Dec. 13 n.13.

We also addressed the construction of the claim limitation “a linker site formed over the light detector, the linker site being treated to affix the biomolecule to the linker site.” We stated that “[a]lthough the challenged apparatus claims are ‘capable of’ affixing a single biomolecule to the linker site, the structure of the apparatus is not limited to being capable of affixing *only* a single biomolecule to the linker site.” *Id.* at 13–14 (citing claim 30).

In the Response, Patent Owner again argues that the preamble of claim 1 is limited to apparatuses “for identifying a single biomolecule.” *See* Resp. 4–14. Patent Owner also argues that the challenged claims require “a linker site” treated to affix a *single* biomolecule to be identified by the apparatus. *Id.* at 10–11. Patent Owner further argues that the claim phrase “linker site formed over the light detector” at least requires that “the linker site be configured directly or indirectly on top of, upon, or above” the light detector. *Id.* at 14–16.

We discuss below each of these claim terms or phrases and the parties’ arguments with respect to the same.

1. Preamble Effect – Identifying a Single Biomolecule

Patent Owner argues that the preamble of claim 1 limits the claims to identifying a single biomolecule. *See* Resp. 4–16. Patent Owner contrasts an apparatus for identifying a single biomolecule with an apparatus for identifying multiple copies of a biomolecule in an ensemble. *Id.* at 9–10 (citing Ex. 2008 ¶ 43). First, Patent Owner argues that the preamble provides antecedent basis for “the biomolecule” referenced in the bodies of the claims. *Id.* at 4–6 (citing *CW Zumbiel Co., Inc. v. Kappos*, 702 F.3d 1371, 1385 (Fed. Cir. 2012)). By relying on the preamble for antecedent basis, Patent Owner argues that the claims use both the preambles and bodies of the claims to define the invention itself. *See id.* at 5–6 (citing *Bell Comms. v. Vitalink Comms. Corp.*, 55 F.3d 615, 620 (Fed. Cir. 1995)).

Second, Patent Owner argues that the preambles recite a fundamental characteristic of the invention, i.e., identifying a single biomolecule. Resp. 6–9 (citing *On Demand Machine Corp. v. Ingram Indus., Inc.*, 442 F.3d 1331, 1343 (Fed. Cir. 2006) (finding the preamble at issue there limited the claims because it stated the framework of the invention and was fundamental to the invention)).

Patent Owner argues that, similar to *On Demand*, “the specification of the ’441 patent describes identifying a single biomolecule as fundamental to the invention.” Resp. 7. Specifically, Patent Owner argues that “[a]ll of the apparatuses in the figures of the ’441 patent are structures for identifying a single biomolecule,” and all of the Examples “describe apparatuses for detecting a single biomolecule and methods for making and using them, including for sequencing a nucleic acid.” *Id.* at 7–8 (citing Ex. 2008 ¶¶ 37–38). Patent Owner argues that a person of ordinary skill in the art would

have understood these disclosures, combined with the Summary and Detailed Description of the Specification, “to communicate that an apparatus for identifying a single biomolecule is fundamental to the invention.” *Id.* at 8 (citing Ex. 2008 ¶ 39).

Patent Owner also asserts that a “single” biomolecule as used in the ’441 patent and required by the claims “is an exclusive word referring to a count of only one.” Resp. 9. Patent Owner concludes that “in view of the specification and the claims, the plain and ordinary meaning of ‘[a]n apparatus for identifying a single biomolecule’ to a POSA is a structure for identifying an individual biomolecule, as opposed to, for example, multiple copies of a biomolecule in an ensemble.” *Id.* (citing Ex. 2008 ¶¶ 40–43; Ex. 1001, 2:16–18, 2:30–31, 4:2–4, 5:49–50, 12:36–58, 15:11–13).

Petitioner responds that the preamble recites “identifying a single molecule,” and not “detecting a signal from only a single molecule.” Reply 2–3. Petitioner asserts that identifying a single molecule can be performed by copying a single target molecule and analyzing the signal from a collection of identical molecules. *See id.* Petitioner asserts that Patent Owner’s expert confirms this interpretation of “identify” with respect to “the well-known Illumina sequencing platform,” which analyzes signal “from a collection of identical molecules that have been amplified on a surface.” *Id.* at 3.

Petitioner asserts that the dependent claims also support its interpretation. *See* Reply 4–6. Specifically, dependent claim 26, recites “wherein the nucleic acid is amplified at the linker site before nucleic acid sequencing.” *Id.* at 4. Petitioner asserts that amplifying the nucleic acid “expressly contemplates detecting a signal from *multiple* molecules of the

same species.” *Id.* (citing Ex. 1003 ¶ 60). Petitioner also refers to claim 30, which recites “affixing **one or more** biomolecule to the linker site of the apparatus of claim 1.” *Id.* at 5.

Petitioner further contends that the Specification, prosecution history, and extrinsic evidence support its claim interpretation. Reply 6–9. First, Petitioner argues that the ’441 patent cites to multiple publications for describing “Sequencing Modalities.” *Id.* at 6–8 (citing Ex. 1001, 12:37–40, 13:2–4). Petitioner contends that Dr. Harris acknowledges these references disclose “techniques in which one detects signal from multiple molecules of the same species.” *Id.* at 6–8 (citing Ex. 1058, 86:16–25, 79:14–80:10, 83:2–84:15, 88:12–17, 77:9–24 (Harris Deposition)).

Second, Petitioner contends that the ’441 patent claims the benefit of provisional application No. 61/036,652, and the provisional application “expressly claims a process ‘wherein a amplification process can be included after said nucleic acid binded to said binding site.’” *Id.* at 8 (citing Ex. 1056 at 40). Finally, Petitioner contends that the extrinsic evidence supports its position because of Dr. Harris’ previous work at Helicos. *Id.* at 8–9 (citing Ex. 2008 ¶¶ 8–11). Specifically, Petitioner asserts that Helicos previously described an alleged infringing “Illumina platform as carrying out ‘single molecule sequencing,’” despite the Illumina platform sequencing multiple copies of a molecule together. *See id.* at 9 (citing Ex. 1054 ¶¶ 39, 137; Ex. 1058, 107:6–9).

In construing claim terms, we start with the language of the claims themselves. *See Phillips*, 415 F.3d at 1314 (stating “the claims themselves provide substantial guidance as to the meaning of particular claim terms”). A preamble to a claim should be construed as limiting the scope of the claim

“[i]f the claim preamble, when read in the context of the entire claim, recites limitations of the claim, or, if the claim preamble is ‘necessary to give life, meaning, and vitality’ to the claim.” *Pitney Bowes, Inc. v. Hewlett-Packard Co.* 182 F.3d 1298, 1305 (Fed. Cir. 1999). The Federal Circuit has also stated:

[D]ependence on a particular disputed preamble phrase for antecedent basis may limit claim scope because it indicates a reliance on both the preamble and claim body to define the claimed invention. Likewise, then the preamble is essential to understand limitations or terms in the claim body, the preamble limits claim scope.

Catalina Marketing Internat’l v. Coolsavings.com, Inc., 289 F.3d 801, 808 (Fed. Cir. 2002) (citing *Bell*, 55 F.3d at 620; *Pitney Bowes*, 182 F.3d at 1306).

Here, we find that the preamble reciting “a single biomolecule” for claim 1 provides antecedent basis for the claim term “the biomolecule” that is affixed to a linker site and recited in the body of claim 1. *See* Ex. 1001, 26:11–18.

Claims must also be read in view of the specification of which they are an integral part. *See Phillips*, 415 F.3d at 1315. The Specification of the ’441 patent, like the language of the preamble itself, repeatedly refers to the detection apparatus as identifying a “single” biomolecule. *See, e.g.*, Ex. 1001, 1:67–2:6 (stating devices should be capable of sequencing *single* molecules), 2:16–17 (stating each sequencing reaction uses a *single* molecule as the template), 3:55–4:16 (stating each optical detection apparatus 20 may be operated independently to detect and identify a *single* biomolecule, and by integrating a huge number of optical detection apparatuses 20 on substrate 10, a huge number of *single* biomolecules can be

detected and identified in parallel). Accordingly, both the claim language and the Specification support Patent Owner's position that the preamble is limiting.

We find Petitioner's argument that the preamble is not limiting and that one can "identify" a single molecule from a signal from multiple copies of the same molecule is inapposite when the challenged claims are read in light of the Specification of the '441 patent. *See* Reply 2–4. Although such a method for identification of a molecule exists and, as Petitioner identifies, is done by the Illumina sequencing platform, that is not what is described in the Specification of the '441 patent and the claims at issue here. *See Phillips*, 415 F.3d at 1317 ("However, while extrinsic evidence 'can shed useful light on the relevant art,' we have explained that it is 'less significant than the intrinsic record in determining "the legally operative meaning of claim language.'") (quoting *C.R. Bard, Inc. v. U.S. Surgical Corp.*, 388 F.3d 858, 862 (Fed. Cir. 2004)).

The Specification of the '441 patent describes a bioassay system in which a very large number of the apparatuses described and claimed in the '441 patent, in some embodiments over 10,000, are used in parallel "to determine, for example, the sequence of a genome or the profile of expressed genes in a tissue sample with high throughput." Ex. 1001, 3:58–65. In such a bioassay system, the Specification indicates that "[e]ach optical detection apparatus may sense the existence of *a* fluorophore on the *single* molecule by detecting photons emitted from the fluorophore." *Id.* at 3:60–61 (emphasis added). The Specification of the '441 patent contemplates running myriad optical detection apparatuses in parallel to detect a *single* or individual biomolecule in each such apparatus. The fact

that other “Sequencing Modalities” are mentioned that may be used with the claimed apparatus of the ’441 patent, and which involve the detection of signal from multiple molecules of the same species, does not undercut this fulsome description of what is disclosed in the ’441 patent. *See* Reply 6–8.

Petitioner also points to dependent claims 26 and 30 as supporting its view that the challenged claims encompass detection of signal from multiple biomolecules of a particular species at a linker site. Reply 5–6. Claim 26 depends from claim 16, which claims a method of sequencing *a* nucleic acid, comprising the steps of affixing *one* nucleic acid molecule to the linker site of the apparatus of claim 1 and performing nucleic acid sequencing of that nucleic acid molecule. Ex. 1001, 26:59–64. Claim 26 adds a step to the method of claim 16 “wherein the nucleic acid is amplified at the linker site before nucleic acid sequencing.” *Id.* at 27:20–21. The term “amplified” is used nowhere else in the ’441 patent.

Dr. Neikirk opines that claim 26 “expressly contemplates detecting a signal from *multiple* molecules of the same species.” Reply 4 (citing Ex. 1003 ¶ 61). Dr. Neikirk opines:

Those of skill in the art understand that this [amplification of the nucleic acid at the linker site before sequencing] refers to a process in which a single molecule is first located at the linker site and then multiple copies of that molecule are generated. Thus, in this case, the signal is detected from multiple molecules of the same species.

Id.

We do not find Dr. Neikirk’s explanation here to be persuasive. Dr. Neikirk, who is admittedly not well-versed in molecular sequencing,⁷ does not explain how “amplification at the linker site” in the context of

⁷ Ex. 2009, 89:8–11; 30:22–25; 37:21–22; 38:14–15.

claim 26 in which only one nucleic acid molecule has been affixed to the linker site would mean that a signal is detected from multiple molecules of the same species. In particular, even if multiple copies of the molecule are made through amplification, claim 26 (by dependency on claim 16) recites that only “one nucleic acid molecule” is affixed to the linker site. There is no further indication in the claim language concerning what is done with the amplified nucleic acid molecules.

Dr. Neikirk’s opinion assumes that the molecules made through amplification must be affixed to the same linker site as the original nucleic acid molecule and detected, an assumption that is not supported by the record evidence. Moreover, Patent Owner explains how the claimed amplification means that the apparatus is still only detecting signal from the single affixed molecule. Resp. 10 (“But such amplification does not imply that multiple copies of a nucleic acid molecule are then affixed to one linker site for detection as an ensemble—particularly given that claim 26 depends from claim 16, which expressly requires performing nucleic acid sequencing of ‘one nucleic acid molecule’ affixed to the linker of the apparatus of claim 1.”) (citing Ex. 2008 ¶¶ 46–48). We credit Patent Owner’s interpretation over Petitioner’s because it more closely aligns with the surrounding claim language and Specification, as explained above.

Petitioner states that claim 30, which recites “affixing *one or more* biomolecule to the linker site of the apparatus of claim 1,” expressly contemplates multiple molecules at the linker site. Reply 5. Petitioner does not cite to any testimony from Dr. Neikirk for this argument.

We agree that claim 30 contemplates using the apparatus of claim 1 in a method where more than one molecule is affixed, but that does not mean

that the apparatus of claim 1 is incapable of “identifying a single biomolecule,” as its preamble recites. Indeed, the fact that Patent Owner uses the phrase “a single biomolecule” in claim 1, but “one or more biomolecule” in claim 30, suggests their meaning is not coextensive, further evidencing that claim 1 should not be construed to encompass an apparatus that can only identify multiple biomolecules.

Accordingly, we determine that the preamble of claim 1 is limiting, requires an apparatus capable of identifying a single biomolecule, and provides antecedent basis for “the biomolecule” as used in the body of the independent claims, which should be read as “the single biomolecule” introduced in the preamble.

In our Institution Decision, we also addressed the scope of the claims as they relate to the linker site limitations, which we have touched on in our discussion of claim 30 above. These limitations require generally “a linker site formed over the light detector, the linker site being treated to affix the biomolecule to the linker site.” *See* Ex. 1001, 26:14–15. In our Institution Decision, we stated:

The challenged apparatus claims recite a structure that includes a linker site being treated⁸ “to affix the biomolecule” thereto, the apparatus thereby being capable of affixing “the [single]⁹ biomolecule” to the linker site. But how the claimed apparatus is intended to be employed does not differentiate the claimed apparatus from a prior art apparatus if the prior art teaches all the structural elements of the claim. *Hewlett-*

⁸ We noted in our Institution Decision that the Specification does not define the term “treated,” but does state that “[a]ppropriate means for affixing nucleic acids to a solid support are well known in the art.” Ex. 1001, 11:8–9.

⁹ For the purpose of the Institution Decision, we interpreted the claim phrase “to affix *the* biomolecule” as referring to “a single biomolecule.”

Packard Co. v. Bausch & Lomb Inc., 909 F.2d 1464, 1468 (Fed. Cir. 1990) (“[A]pparatus claims cover what a device *is*, not what a device *does*.”). As the Federal Circuit has recently reiterated, this means that a prior art reference “may anticipate or render obvious an apparatus claim—depending on the claim language—if the reference discloses an apparatus that is reasonably capable of operating so as to meet the claim limitations, even if it does not meet the claim limitations in all modes of operation.” *ParkerVision, Inc. v. Qualcomm Inc.*, 903 F.3d 1354, 1361 (Fed. Cir. 2018); *see also* Ex. 1001, 1:67–2:1 (“the devices should be ‘capable of’ sequencing single molecules”).

Although the challenged apparatus claims are “capable of” affixing a single biomolecule to the linker site, the structure of the apparatus is not limited to being capable of affixing *only* a single biomolecule to the linker site. This is evident, at least with respect to claim 1, by the recitation of claim 30, *viz.*, “affixing one or more biomolecule[s] to the linker site of the apparatus of claim 1.” Ex. 1001, 27:32–33; *see also Phillips*, 415 F.3d at 1314 (“the claims themselves provide substantial guidance as to the meaning of particular claim terms”). We further note that all of the challenged apparatus claims include the open ended transitional phrase “comprising,” and thus do not limit the claimed structure to a linker site that is capable of affixing *only* a single biomolecule for identification thereof. *See CIAS, Inc. v. Alliance Gaming Corp.*, 504 F.3d 1356, 1360 (Fed. Cir. 2007) (“In the patent claim context, the term ‘comprising’ is well understood to mean ‘including but not limited to.’”).

Dec. 13–14.

Patent Owner challenges our preliminary determinations concerning how to construe the linker site limitations. Patent Owner asserts that a person of ordinary skill in the art “would have understood that the ‘linker site being treated to affix the [single] biomolecule’ limitation describes a structural configuration, not merely a capability.” Resp. 11 (citing Ex. 2008 ¶ 54). Patent Owner argues further that the transitional phrase “comprising,”

does not nullify the express recitation of identifying a single biomolecule. *Id.* at 12–13 (citing *On Demand*, 442 F.3d at 1343–1344).

Petitioner agrees with our initial assessment set forth above. *See* Reply 15–19. Petitioner states: “The claim language here recites ‘apparatus ***for identifying*** a single biomolecule’ and ‘linker site being treated to affix the biomolecule to the linker site.’ Such language mirrors what the Federal Circuit has found to recite capability as opposed to configuration.” *Id.* at 17 (citing *ParkerVision*, 903 F.3d at 1362).

We agree with Petitioner that our original construction is correct. Examining the claim language is crucial to determine whether it recites a mere capability to carry out a function versus a particular configuration to carry out the function. *See ParkerVision*, 903 F.3d at 1361. As the Federal Circuit stated in *ParkerVision*, “[t]he language used in the claims is critical to deciding on which side of the line the claims fall.” *Id.*

Here, the claims describe the required function of “identifying a single biomolecule” using an apparatus with (1) a substrate having a light detector and (2) a linker site formed over the substrate, being treated to affix the biomolecule to the linker site, with the linker site a distance of less than or equal to 100 micrometers from the light detector. *See* Ex. 1001, 26:11–18. The structure described includes a substrate with a light detector and a linker site over the substrate and no more than 100 micrometers away from the light detector. No specific structure is delineated in the claims for the linker site that would restrict use of the site for only a single biomolecule. The only requirement in the claims is the linker site *being treated* to affix the single biomolecule to it. Such a limitation is more akin to the types of claims that the Federal Circuit has deemed to be claim language reciting a

capability versus a particular configuration. *See Ball Aerosol & Specialty Container, Inc. v. Limited Brands, Inc.*, 555 F.3d 984, 987 (Fed. Cir. 2009) (finding noninfringement where claims recited a configuration in which protrusions needed to be “resting upon” a cover and alleged infringing device was only reasonably capable of such a configuration); *Ericsson, Inc. v. D-Link Sys., Inc.*, 773 F.3d 1201, 1217 (Fed. Cir. 2014) (finding claims at issue recite capability such as “a processor *for arranging information* for transmission . . . which identifies a type of payload information”).

Therefore, we conclude that although the challenged claims requiring an apparatus “identifying a single biomolecule” are “capable of” affixing a single biomolecule to the linker site, we remain persuaded that the structure of the apparatus is not limited to being capable of affixing *only* a single biomolecule to the linker site.

2. *Formed Over*

Patent Owner argues that a person of ordinary skill in the art would have understood “‘a linker site formed over the light detector’ to at least require that the linker site be configured directly or indirectly on top of, upon, or above and, as some dictionaries articulate it, ‘at a higher level or layer than,’ . . . the light detector.” Resp. 14–15 (citing Ex. 2008 ¶¶ 59–60; Ex. 2012; Ex. 2013). Patent Owner argues that Figure 2 of the ’441 patent depicts linker site 220 in a layer of the device above light detector 210. *Id.* at 15–18 (citing Ex. 1001, Fig. 2, 5:33–35; Ex. 2008 ¶¶ 61–62). Patent Owner further argues that Figures 1–4 support this interpretation, where a person of ordinary skill in the art would have understood formed over as meaning “in a layer above.” *Id.* at 16 (citing Ex. 2008 ¶¶ 63–64).

Petitioner contends that “formed over” should not be construed to require either a “layer” or positioning “above.” *See* Reply 9–11. For example, Petitioner asserts that the Specification does not require that the light source be in a layer, and instead discloses embodiments in which “the excitation light source may be *stand alone* from optical detection apparatuses 20 or bioassay substrate 10” or “the excitation light source may be integrated with optical detection apparatuses 20 or bioassay substrate 10.” *Id.* at 10 (citing Ex. 1001, 7:4–9). Petitioner contrasts claim 49 which expressly recites a light emitting layer. *See id.* (citing claim 49). Petitioner further contends that the claims do not recite the term “above,” and that interpreting the “formed over” to require “above” “would exclude from the scope of the claims disclosed embodiments simply if they are turned upside down.” *Id.* at 10–11 (citing Ex. 1058, 142:15–22, 148:14–15, 154:8–14, 154:23–155:6, 155:10–15).

In reviewing the totality of the claim language describing the apparatus for identifying a single biomolecule in light of the Specification of the ’441 patent, it appears that the term “formed over” describes the close proximity of the linker site and the light detector so that the light detector can collect light emitted from the biomolecule and the relative positions of the linker site as over or above the light detector. *See, e.g.*, Ex. 1001, Abst., 2:30–41. For instance, in the description of Figure 2 set forth above, *see supra* Section I.D., optical detection apparatus 20 is described as including a “light detection 210 formed on substrate 10, and a linker site 220 formed over light detector 210” and shows the linker site placed above the light detector. *Id.* at 4:55–57.

The Specification of the '441 patent makes clear that the term “over” means at a higher level vertically or above. For instance, in discussing Figure 3 set forth below, the Specification of the '441 patent states:

Referring to FIG. 3, there is illustrated a sectional view of optical detection apparatus 20 in accordance with one embodiment consistent with the present invention. As shown in FIG. 3, blind sheet 230 is *formed over* light detector 210 and *vertically spaced apart* from light detector 210 by distance H1. Blind sheet 230, which has a thickness T, includes pinhole 235 having a radius R1 (i.e., one-half of diameter D1). In this embodiment, linker site 220 may be *formed in* pinhole 235 to bind with a biomolecule (not shown).

Ex. 1001, 5:65–6:6 (emphases added).

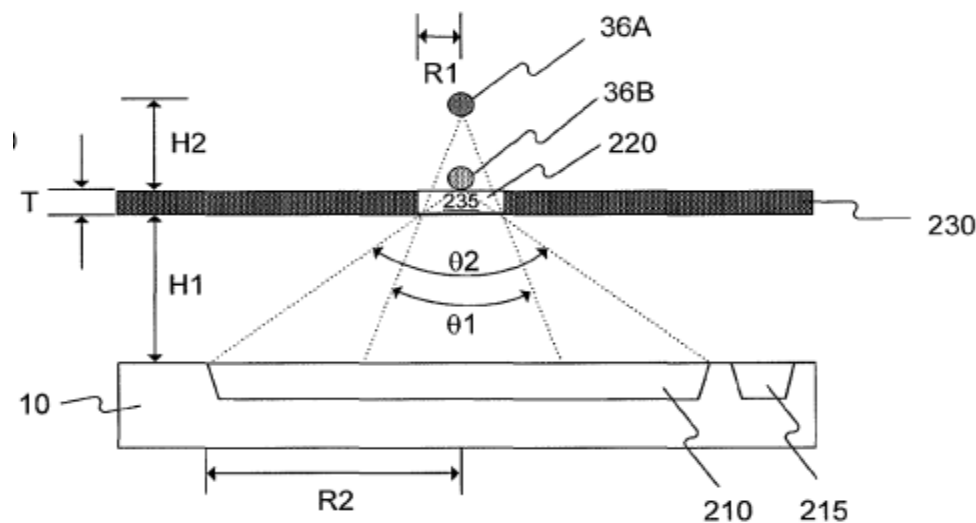


Fig. 3

The Specification of the '441 patent further highlights the meaning of “over” by juxtaposing it with forming “under.” For instance, in describing optical detection apparatus 20, the Specification of the '441 patent makes a distinction between a component “formed over” or “formed under.”

Optical detection apparatus 20 may further include a filter layer 240 (optional) and a microlens 250 (optional) between light detector 210 and blind sheet 230. Although FIG. 2 shows filter layer 240 is *formed over* microlens 250, it is

appreciated that filter layer 240 may be *formed under* microlens 250.

Ex. 1001, 5:31–35. Finally, the Specification of the ’441 patent further highlights that “over” means above in describing a method for manufacturing a bioassay system.

In this embodiment, a filter layer 240 is *formed over* the *upper surface* of light detectors 210 and control circuits 215. Global planarization process is applied to the *upper surface* of light detectors 210 and control circuits 215 before forming filter layer 240. Filter layer 240 includes a plurality of sublayers. In this embodiment, filter layer 240 is formed by first depositing a sublayer having a higher refractive index *over* the planarized *upper surface* of light detectors 210 and control circuits 215.

Id. at 19:47–55.

Therefore, we determine that “formed over” as used in the challenged claims means “placed above.”¹⁰

D. Anticipation by Hassibi

Petitioner asserts that claims 1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36 are anticipated by Hassibi. Pet. 9. Petitioner provides an analysis of how it alleges that each claim limitation is taught by Hassibi. *Id.* at 16–44.

Petitioner relies on the declaration of Dr. Neikirk in support of its positions. *See generally* Ex. 1003. Patent Owner responds by asserting that Hassibi does not disclose several of the limitations of the challenged claims. *See* Resp. 16–46.

¹⁰ We do not see any description that restricts the linker site in a “layer.” Therefore, we do not include this in the construction of the term “formed over.”

1. Hassibi (Ex. 1005)

Hassibi discloses “methods, compositions and apparatus for detecting, [i]dentifying, quantifying and/or sequencing target biomolecules, such as nucleic acids or proteins.” Ex. 1005, Abst. Hassibi also discloses that “the methods may comprise obtaining at least one sample suspected of containing one or more target biomolecules. Where the target of interest is a nucleic acid, it may be captured and/or isolated by a variety of known techniques, such as sequence specific hybridization with one or more capture probes.” *Id.* ¶ 12.

Figure 8 of Hassibi illustrates a detection apparatus and is reproduced below.

FIG. 8

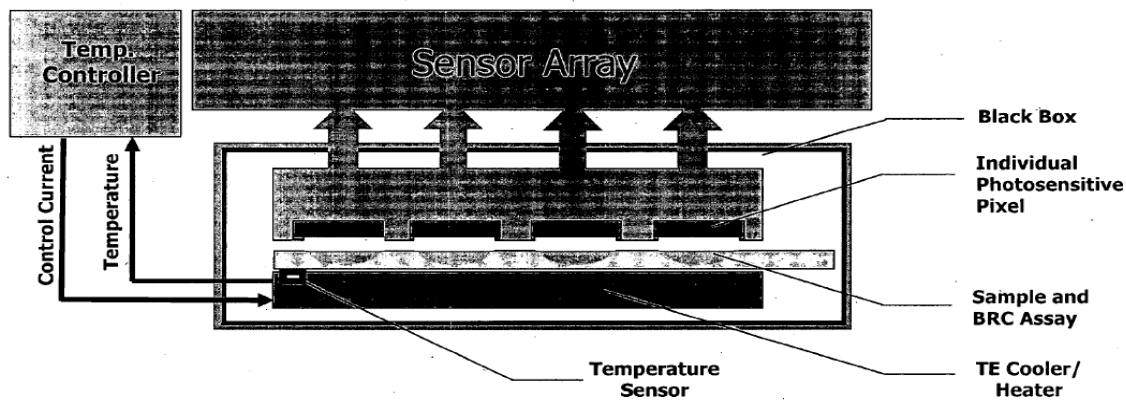


Figure 8 illustrates an apparatus for use with Bioluminescence Regenerative Cycle (“BRC”) detection. Ex. 1005 ¶ 38. As shown above, Figure 8 illustrates reaction chambers for affinity capture of targets, optical coupling devices to convey generated photons to photodetectors, photodetectors to generate a relative photocurrent from incidence photons, and a sensor array to efficiently acquire and measure photocurrent. *Id.*

¶¶ 139–145. Hassibi discloses that the reaction chambers “may contain capture medium to allow target biomolecules (e.g. nucleic acids and/or proteins) to be specifically captured using different types of affinity matrices, functionalized gels and/or probes immobilized on solid surfaces.” *Id.* ¶ 151.

Hassibi discloses that the optical coupling device may collect light generated from the reaction chambers and transfer the light to a photodetector. Ex. 1005 ¶ 163. In embodiments of the invention, “the generated photons may either propagate for a short distance to a [photodetector], or may reach a photodetector substantially in contact with the chamber wall (distance from detector to chamber can vary from 1 μ m to 1 m, but typically would range from 10 μ m to 2 mm).” *Id.* Hassibi further discloses that “[i]n some embodiments, the photodetector can be in close proximity to the chamber and/or even integrated onto the chambers.” *Id.* ¶ 165. Examples of photodetectors include “an array of photodiodes in silicon wafers, where chambers are etched into either the oxide top layers, or the bulk silicon wafer,” or a micro-fluidic chip “put onto the surface of a semiconductor based image sensor (e.g. CMOS or CCD), where the light from each well directly impinges on a photosensitive section of the imager.” *Id.*

Hassibi discloses “the target biomolecule(s) of interest may be captured, immobilized and/or labeled by binding to one or more binding moieties.” *Id.* ¶ 238. Exemplary binding moieties include antibodies and aptamers. *Id.* Other examples include attaching binding moieties, capture probes, or analytes of interest to a surface by covalent or non-covalent interaction, and promoting such attachments with cross-linking reagents. *Id.*

¶ 262. An exemplary method may include coating a surface “with streptavidin or avidin with the subsequent attachment of a biotinylated molecule, such as an antibody or analyte.” *Id.* ¶ 264.

2. Analysis

Petitioner argues that Hassibi discloses the limitations of claim 1, and relies on the Neikirk Declaration as support. Pet. 16–28 (citing Ex. 1003 ¶¶ 60–82). Patent Owner argues that Hassibi does not disclose several of the limitations of claims challenged in this ground. Resp. 11–22.

We determine that Hassibi fails to disclose an apparatus that is capable of identifying a single molecule. Therefore, we find that Petitioner has failed to show by a preponderance of the evidence that Hassibi teaches “an apparatus for identifying a single biomolecule.”

Although Petitioner argues that the preamble of claim 1 is not limiting, it also argues that Hassibi discloses “[a]n apparatus for identifying a single biomolecule” by disclosing a method of obtaining samples containing “one or more target biomolecules.” See Pet. 16–18 (citing Ex. 1005, Abst., ¶¶ 12, 64, claim 1; Ex. 1003 ¶¶ 62–64); see Ex. 1005 ¶ 64 (“‘Targets’ are not limited to single molecules or atoms”).

Specifically, Petitioner asserts the following to establish that Hassibi teaches “an apparatus for identifying a single biomolecule.”

Hassibi discloses that its “invention concerns methods, compositions and apparatus for detecting[,] [i]dentifying, quantifying and/or sequencing target biomolecules, such as nucleic acids or proteins.” EX1005 at Abstract. Hassibi discloses that the “target” biological material may comprise a single molecule. *Id.* ¶ 12 (“In certain embodiments of the invention, the methods may comprise obtaining at least one sample suspected of containing *one* or more target biomolecules.”).

The claims in Hassibi further disclose the identification of a single biomolecule. For instance, Claim 1 recites “obtaining at least one sample suspected of containing **one** or more target biomolecules... detecting the target biomolecules in the sample.” *Id.* at Claim 1.

Consistent with this, Claim 5 recites that the target biomolecule can be “a nucleic acid,” while Claim 6 recites that the target nucleic acid can be “amplified.” *See id.* at Claims 5-6. Those of skill in the art understand that a nucleic acid that has been “amplified” corresponds to a nucleic acid that has been copied. EX1003 ¶ 64. As if this was not clear enough, Hassibi further clarifies that the term “target” is “not limited to **single molecules** or atoms.” EX1005 ¶ 64. Thus, in Hassibi, one may detect a signal from either a single biomolecule, or one can detect a signal from multiple copies of the same biomolecule. Both of these options constitute disclosure of the preamble of Claim 1.

Pet. 17–18.

Patent Owner responds that “Hassibi’s specification discloses methods for detecting multiple copies of a biomolecule together,” as opposed to identifying a single biomolecule. Resp. 19 (citing Ex. 2008 ¶ 82). Specifically, Patent Owner argues that Hassibi discloses a BRC assay with a sensitivity of detection as low as 0.1 attomoles (amol). *Id.* at 18–19 (citing Ex. 2008 ¶¶ 72–78; Ex. 1005 ¶¶ 9, 17–19, 25, 69). Patent Owner argues that because there are 60,000 molecules in 0.1 amol, Hassibi’s apparatus is only able to detect biomolecules where there are more than 60,000 molecules in the sample. *Id.* at 19 (citing Ex. 2008 ¶ 82; Ex. 2009, 88:5–89:1). Patent Owner argues that “[o]ver and over again, Hassibi discloses an apparatus that analyzes huge numbers of biomolecules together, not an apparatus for identifying one biomolecule individually, as claimed.” *Id.* at 21. Moreover, Patent Owner argues that “Hassibi does not describe

any particular methods of sequencing, let alone methods of sequencing a single biomolecule.” *Id.* at 23 (citing Ex. 2008 ¶ 100).

With respect to Hassibi’s disclosure of identifying one or more target biomolecules, Patent Owner argues that Hassibi refers to types of biomolecules, not total number of biomolecules. *See* Resp. 24–28. Patent Owner argues that Hassibi describes various examples of apparatuses for site-specific capture of different targets in the same sample. *Id.* at 25 (citing Ex. 1005 ¶¶ 138–139, 154, 269, 382–384). Patent Owner argues that thus Hassibi refers “to a sample containing one or more species of biomolecule to be targeted, not a total of one or more biomolecules of a particular species in the sample.” *Id.* at 26 (citing Ex. 2008 ¶¶ 113–117).

The question that we need to answer is whether Hassibi teaches an apparatus that is capable of identifying a single biomolecule. We agree with Patent Owner that Hassibi does not. The disclosures in Hassibi that Petitioner points us to do not teach an apparatus that is capable of identifying a single biomolecule.

Petitioner points us to Hassibi’s description that the “target” encompasses a single molecule. Pet. 16–18, 23–26; *see* Ex. 1003 ¶¶ 60–64. For instance, the Abstract describes Hassibi’s invention as “methods, compositions and apparatus for detecting[, i]dentifying, quantifying and/or sequencing target biomolecules, such as nucleic acids or proteins.” Pet. 17 (citing Ex. 1005, Abst.) Petitioner links this statement to the described “obtaining” of a target biological material or sample that is “suspected of containing *one* or more target biomolecules.” *Id.* at 17–18 (citing Ex. 1005 ¶ 12). From these two statements and claim language from claims 1, 5, and 6 of Hassibi, Petitioner concludes that Hassibi teaches “identifying a single

biomolecule.” Pet. 17–18. For instance, Petitioner asserts, claim 1 also describes “obtaining at least one sample suspected of containing *one* or more target biomolecules” and then “detecting the target biomolecules in the sample;” claim 5 describes that the target molecule can be “a nucleic acid;” and claim 6 states that this target molecule can be “amplified.” *Id.* at 18.

Petitioner’s citations to Hassibi as outlined above, however, do not provide a sufficient link between a single target molecule and the identification of such a molecule in view of Hassibi’s teachings regarding how molecules are identified in its apparatus. Hassibi’s Abstract describes:

The presence of target molecules in samples may be detected by a variety of enzymatic processes that generate a detectable product, such as pyrophosphate (PPi) or ATP. In preferred embodiments of the invention, the product is detected by a bioluminescence regenerative cycle (BRC), utilizing luciferase mediated bioluminescence.

Ex. 1005, Abst., ¶ 15.

Dr. Harris explains bioluminescence regenerative cycle or BRC as disclosed in Hassibi “uses a coupled reaction where pyrophosphate is generated by an enzyme-catalyzed process, such as nucleic acid polymerization,” which includes the addition of individual deoxyribose nucleoside triphosphates (dNTPs) or nucleoside triphosphates (NTPs) to a chain of DNA or RNA, respectively. *See* Ex. 2008 ¶¶ 71–75. Referring to Hassibi, Dr. Harris explains that during incorporation of dNTP or NTP, two of the three phosphate groups in the triphosphate part of the dNTP or NTP molecule break off together and are known as pyrophosphate or PPi. *Id.* ¶¶ 74–75.

Consistent with Dr. Harris’ testimony, Hassibi explains BRC analysis as follows:

In certain embodiments of the invention, the pyrophosphate producing reaction is allowed to proceed to completion before BRC analysis. Once the reaction is complete, the pyrophosphate is reacted with APS (adenosine 5'-phosphosulfate) in the presence of ATP Sulfurylase to produce ATP and Sulphate. The ATP is reacted with oxygen and luciferin in the presence of luciferase to yield oxyluciferin, AMP and pyrophosphate. The PPi may react again with APS to regenerate ATP. For each molecule of pyrophosphate that is cycled through BRC, a photon of light is emitted with a quantum efficiency of 0.88 and one molecule of pyrophosphate is regenerated. Because of the relative kinetic rates of luciferase and ATP sulfurylase, a steady state is reached in which the concentrations of ATP and pyrophosphate and the level of photon output remain relatively constant over an extended period of time. The number of photons may be counted (integrated) over a time interval to determine the number of target nucleic acids in the Sample. The very high sensitivity of BRC is related in part to the integration of light output over time, in contrast to other methods that measure light output at a single time point or at a small number of fixed time points. The ability to vary the length of time over which photon integration occurs also contributes to the very high and controllable dynamic range for nucleic acid molecule quantification, with a sensitivity of detection as low as 0.1 attomoles (amol). Increasing the length of integration also significantly reduces detection noise.

Ex. 1005 ¶ 17.

As Dr. Harris explains, there are “more than 60,000 molecules in 0.1 attamoles” meaning that the sensitivity of the BRC analysis described in Hassibi does not identify a single biomolecule because it lacks the sensitivity for the capability to do so. *See* Ex. 2008 ¶ 82. As Dr. Harris concludes: “Thus, far from identifying a single biomolecule, Hassibi’s specification discloses methods for detecting multiple copies of a biomolecule together. Indeed, Hassibi is stating that a sample must contain more than 60,000

molecules to be detected using its BRC assay.” *Id.* Dr. Harris also identifies several examples in Hassibi that bear out this interpretation of how that apparatus in Hassibi works. *See id.* ¶¶ 83–93 (describing Example 1 with a similar sensitivity of 0.1 attamole, FIG. 10 describing BRC analysis of a 6 billion molecule sample, and Example 2 with an over 6 million molecules sample analyzed by BRC) (citing Ex. 1005 ¶¶ 321, 325). Dr. Harris concludes: “Again and again, Hassibi discloses an apparatus that analyzes huge numbers of biomolecules together, not an apparatus for identifying one biomolecule individually. Nowhere does Hassibi disclose an apparatus or method for analyzing one biomolecule alone. And nowhere does Hassibi disclose single-molecule detection sensitivity.” Ex. 2008 ¶¶ 92–93; *see* Pet. 17–22.

Dr. Harris’ testimony and the description of BRC in Hassibi itself demonstrate that Petitioner’s reliance on claim 1 of Hassibi to try to show that it teaches an apparatus capable of identifying a single biomolecule is misplaced. Petitioner only cites to the first and last steps of the method of claim 1 that starts with “obtaining at least one sample suspected of containing one or more biomolecules” and ends with “detecting the target biomolecules in the sample.” The intervening steps, however, of generating PPI, using the PPI to produce light, and accumulating the number of photons produced over a time interval, indicate from the discussion of BRC in Hassibi and Dr. Harris’ explanation set forth above that many more than a single target biomolecule is in the obtained sample.

We find persuasive Dr. Harris’ analysis of Hassibi’s teachings set forth above describing BRC analysis and its sensitivity capability. Petitioner did not file a rebuttal declaration from Dr. Neikirk addressing Dr. Harris’

testimony here. We credit Dr. Harris’ testimony over Dr. Neikirk’s testimony that, as explained below, does not address the complete disclosure of Hassibi.

Dr. Harris also addressed the statements in Hassibi that Petitioner relies on concerning “obtaining at least one sample suspected of containing *one* or more target biomolecules.” *See* Pet. 24–26; Ex. 2008 ¶¶ 112–117. Dr. Harris points out that Dr. Neikirk’s reliance on this statement to establish that Hassibi teaches an apparatus for identifying a single molecule “assumes that nucleic acid amplification typically begins with only one piece of nucleic acid to amplify—i.e., one template molecule. That assumption is incorrect.” Ex. 2008 ¶ 118. Dr. Harris relies on several examples in Hassibi to show that amplification performed in these examples was done on a large number of molecules. *Id.* ¶¶ 120–122. Dr. Harris concludes:

Because a reference to nucleic acid amplification does not imply starting the amplification process with only a single piece of DNA, as confirmed by Hassibi itself, the reference in claim 6 to “[t]he method of claim 5, wherein the target nucleic acid is amplified” does not suggest that Claim 5 is referring to a single molecule, contrary to Dr. Neikirk’s suggestion.

Rather, a person having ordinary skill in the art would have understood claim 5 of Hassibi to specify that the target is a species of nucleic acid or a species of oligonucleotide, not a single biomolecule.

Id. ¶¶ 123–124. We find persuasive Dr. Harris’ detailed assessment of the disclosure in Hassibi and his conclusion that the target molecule of claim 5 is a species of nucleic acid or oligonucleotide and not a single biomolecule as evidenced by multiple examples disclosed in Hassibi.

We also find that the record supports Dr. Harris’ testimony that the definition of “target” in Hassibi does not indicate that a target is a single

molecule, but rather described the types of “targets” that are contemplated. *See* Ex. 2008 ¶¶ 125–126. For instance, Hassibi states that a target may be “any compound, molecule or aggregate of interest for detection.” Ex. 1005 ¶ 64.

In the definition section of Hassibi, it states:

As used herein, the terms “analyte” “biomolecule” and “target” mean any compound, molecule or aggregate of interest for detection. . . . “Targets” are not limited to single molecules or atoms, but may also comprise complex aggregates, such as a virus, bacterium, Salmonella, Streptococcus, Legionella, *E. coli*, Giardia, Cryptosporidium, Rickettsia, spore, mold, yeast, algae, amoebae, dinoflagellate, unicellular organism, pathogen or cell.

Ex. 1005 ¶ 64.

Based on this disclosure, we are persuaded that “a person having ordinary skill in the art would have understood that Hassibi is merely explaining that targets can be unimolecular species (e.g. a piece of DNA) or multi-molecular species (e.g., a complex aggregate of molecules such as a virus.)” Ex. 2008 ¶ 126. We credit this interpretation by Dr. Harris over that offered by Petitioner because it better aligns with the rest of the disclosure in Hassibi and a skilled artisan’s understanding of BRC, as noted above.

Finally, Petitioner points to Figure 4 and 7 of Hassibi to demonstrate “the linker site being treated to affix the [single] biomolecule to the linker site.” *See* Pet. 23–26. Petitioner states: “FIGS. 4 and 7 of Hassibi illustrate a capture probe binding a target molecule to the linker site. FIG. 4 ‘illustrates an exemplary embodiment wherein a capture probe is used to bind to the target molecule [and] attach it to a solid substrate.’ Likewise, FIG. 7 of Hassibi depicts a further embodiment where the target DNA

molecule is captured at the linker site (at 2a in the figure).” *Id.* at 23–24 (citing Ex. 1005 ¶¶ 101, 118). Figure 4 and 7 are set forth below.

FIG. 4

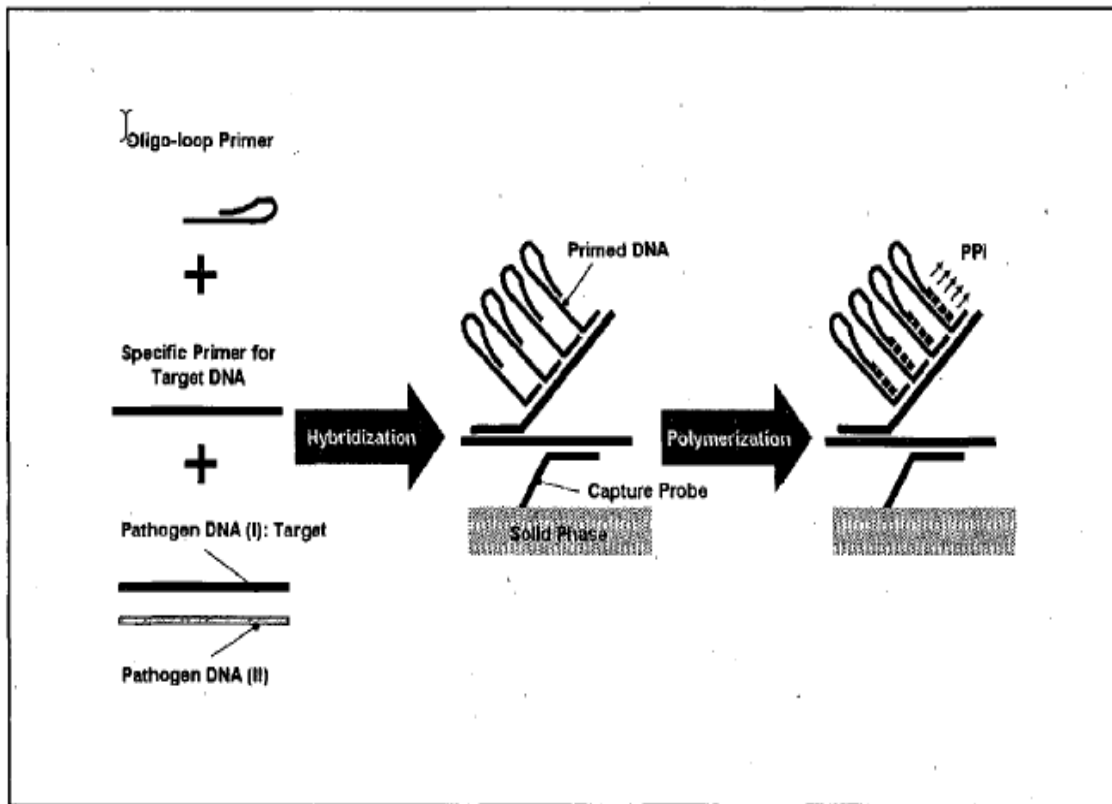


FIG. 7

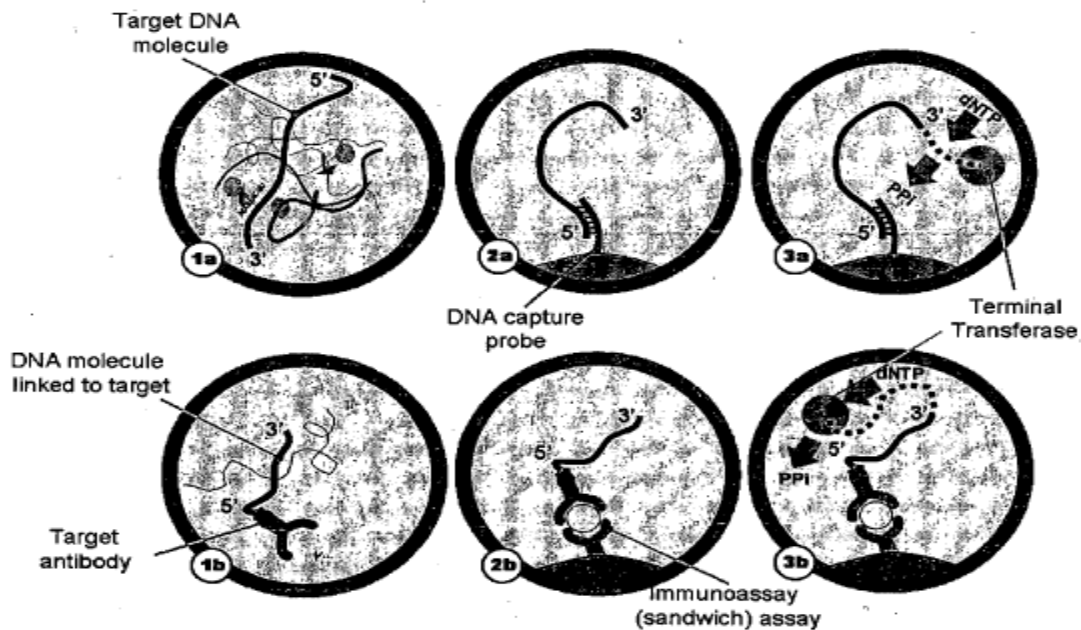


Figure 4 set forth above is described as illustrating “an *exemplary* method for branched chain BRC assay,” and Figure 7 illustrates “*exemplary* methods of terminal transferase based assays.” Ex. 1005 ¶¶ 34, 37 (emphases added).

Petitioner’s reliance on Figures 4 and 7 is unpersuasive. We credit Dr. Harris’ testimony that Figures 4 and 7 in Hassibi are merely magnified examples to show the mechanism of action for the branched chain BRC and terminal transferase based assays. See Ex. 2008 ¶ 143. For instance, Hassibi describes Figure 4 as illustrating “an exemplary embodiment wherein a capture probe is used to bind to the target molecule [and] attach it to a solid substrate. This branching method can potentially generate *thousands* of pyrophosphate molecules per target biomolecule.” Ex. 1005 ¶

101. Figure 7 is specifically described as “[a] general scheme for methods of use of terminal transferase for target molecule detection and/or quantitation.” *Id.* ¶ 117. Therefore, from the described exemplary nature of Figures 4 and 7 in Hassibi, we agree with Dr. Harris’ conclusion that “[o]n these zoomed-in scales, a person of ordinary skill in the art would not have expected to see in the drawings any of the other biomolecules that are present at each alleged linker site of Hassibi.” *Id.*

For all of these reasons set forth above, we find that Hassibi does not teach an apparatus for identifying a single biomolecule as required by all challenged claims. Therefore, we determine that Petitioner has failed to show by a preponderance of the evidence that claims 1, 7, 10–16, 18–19, 21–22, 24, 27, and 30–36 are anticipated by Hassibi.

E. Remaining Grounds Involving Hassibi

Petitioner asserts that claims 1, 7, 10–16, 18–19, 21–22, 24, 27, and 30–36 would have been obvious over Hassibi in view of the skill of a POSA. Pet. 9. In this ground, Petitioner offers no additional argument or evidence that Hassibi teaches an apparatus for identifying a single biomolecule. *See Id.* at 44–51. Therefore, for the same reasons as set forth above, *see* Section II.D., we determine that Petitioner has failed to show by a preponderance of the evidence that any of claims 1, 7, 10–16, 18–19, 21–22, 24, 27, or 30–36 would have been obvious over Hassibi in view of the skill of a POSA.

Petitioner also challenges additional claims or asserts additional combinations of art in the following additional grounds based on Hassibi: (1) Claims 17 and 20 would have been obvious over Hassibi and Korlach; (2) Claims 28 and 29 would have been obvious over Hassibi in view of the skill of a POSA; and (3) Claims 21, 22, and 24 would have been obvious

over Hassibi and Cheeseman. Pet. 9. To challenge these claims in these grounds, Petitioner does not offer any additional argument or evidence that Hassibi, Korlach, or Cheeseman teaches an apparatus for identifying a single biomolecule. *See id.* at 51–63. Therefore, for the same reasons set forth above, *see* Section II.D., we determine that Petitioner has failed to show by a preponderance of the evidence that claims 17 or 20 would have been obvious over Hassibi and Korlach, that claims 28 or 29 would have been obvious over Hassibi in view of the skill of a POSA, or that claims 21, 22, or 24 would have been obvious over Hassibi and Cheeseman.

F. Anticipation by or Obviousness over Blumenfeld

Petitioner asserts that claims 1, 2, and 6 are unpatentable under 35 U.S.C. § 102 as anticipated by Blumenfeld. Pet. 63–74 (citing Ex. 1003 ¶¶ 192–237; Ex. 1007 ¶¶ 2, 3, 17, 25, 46, 66, 74, 76, 77, 81, 83, 86, 88, 100, 111, 136, claims 17, 38, 49, Figs. 1, 2, 3B, 5, 6, 7, 8A, 8B, 9A, 12–14). Alternatively, Petitioner asserts that claims 1, 2, and 6 are unpatentable under 35 U.S.C. § 103 as obvious over Blumenfeld. *Id.* at 74–76.

1. Blumenfeld (Ex. 1007)

Blumenfeld discloses a system for imaging and/or mapping biological samples using electronic light detectors. Ex. 1007 ¶ 2. The system may include “a positioning apparatus for providing a biological sample in a sampling position.” *Id.* ¶ 15. Blumenfeld also discloses that “[t]he system further includes an electronic light detector array with the electronic light detector array including a plurality of detector pixels located at particular detector pixel addresses. The plurality of detector pixels face the first side of the biological sample to receive light therefrom.” *Id.*

Figure 2 of Blumenfeld depicts an imaging apparatus and is reproduced below.

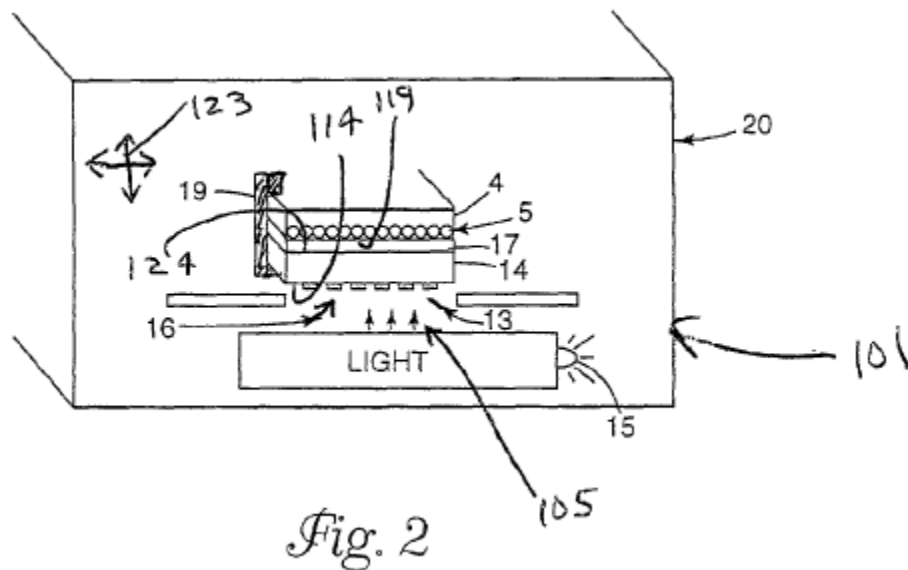


Figure 2 illustrates an electronic light detector array detection system for acquiring and reconstructing light representative of DNA spots on a DNA chip. Ex. 1007 ¶ 30.

As shown above, image detection apparatus 101 includes DNA chip 14, electronic light detector array 4, and optical filter 17, held in contact with each other by clamping system 19. *Id.* ¶ 76. Light detector 4 includes detector pixels 5 for sensing light. *Id.* ¶ 79. Light source 15 provides ultraviolet or blue light 105 that excites light emitted by DNA chip 14 to acquire image 13 representative of sensed light associated with a biological sample. *Id.* ¶ 76. Blumenfeld discloses that “the sampling distance (or the traveling distance for light emanating from one or more portions of the sample to reach the detector pixels 5, i.e., the light path) for use in detection of spots of polynucleic acid or other biological material is” preferably less than 75 microns, most preferably less than 15 microns. *Id.* ¶ 81.

Blumenfeld discloses that DNA chip 14 has flat sample surface 114 with sequences of nucleic acids immobilized thereto, with each sequence being immobilized to a particular chip address. Ex. 1007 ¶ 86. Sample surface 114 of chip 14 is set at a defined position relative to electronic light detector array 4 such that light leaving a chip address on chip 14 is substantially directed through the light path that includes the biological material holding structure, for example a DNA chip substrate, onto at least one detector pixel 5 with an address that is correlated to the chip address. *Id.* Blumenfeld discloses that DNA may be attached to substrates, e.g., forming DNA chips, that pass light (e.g., light transparent DNA chips) “in a variety of manners known to those skilled in the art.” *Id.* ¶ 88. Blumenfeld further discloses that, for example, “glass or quartz may be treated with silanes to create carboxyl or amine groups that may be used in further chemical reactions for immobilizing DNA.” *Id.*

Blumenfeld discloses “[c]hip dimensions, the number of sites of DNA deposition (sometimes termed ‘addresses’) per chip, and the width of the DNA spot per ‘address’ are dependent upon the technologies employed for deposition.” Ex. 1007 ¶ 4. Known techniques produce spots with diameters of 50–300 μm , and as small as 1 micron. *Id.* Blumenfeld also discloses that “[t]echnologies for making such chips are known to those skilled in the art and are described, for instance, in U.S. Pat. Nos. 5,925,525; 5,919,523; 5,837,832; and 5,744,305; which are all incorporated herein by reference.” *Id.*

Figure 5 of Blumenfeld is reproduced below.

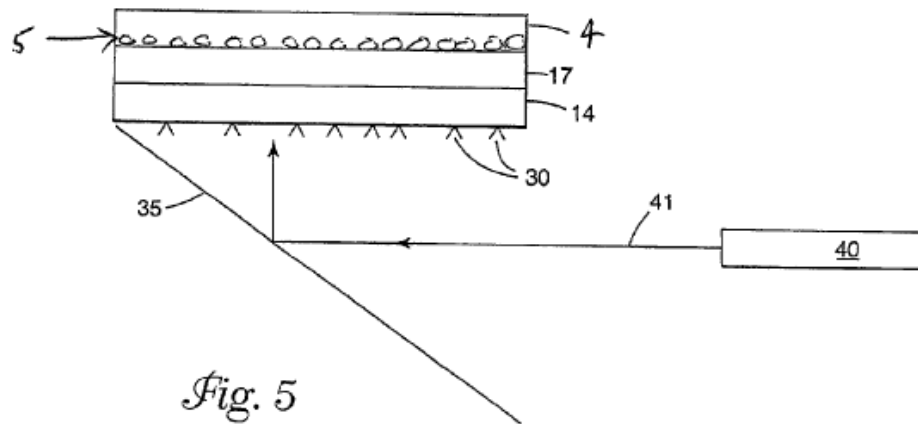


Figure 5 illustrates an electronic light detector array detection system with a laser-light source. Ex. 1007 ¶ 33.

As shown in Figure 5 above, laser 40 may generate laser light 41 reflected off of mirror 35 onto labeled DNA 30 hybridized and immobilized on DNA chip 14. Ex. 1007 ¶ 100. By incorporating fluorescent labels, hybridized DNA 30 fluoresces and emits fluorescent light upon excitation by laser light 41. *Id.* The fluorescent light, but not the laser light, selectively passes through filter 17 and is directly mapped onto detector pixels 5 of detector 4. *Id.* Blumenfeld also discloses that “descriptions of imaging fluorescently immobilized biomolecules and analysis of the images are set forth in U.S. Pat. Nos. 5,874,219 . . . incorporated herein by reference.” *Id.* ¶ 5.

2. Analysis

The Petition includes a limitation-by-limitation comparison of claims 1, 2, and 6 to the disclosure of Blumenfeld, relying on Dr. Neikirk’s Declaration as support. Pet. 63–71 (citing Ex. 1003 ¶¶ 192–219).

Petitioner argues, for example, that Blumenfeld expressly teaches that, referring to Figure 5, individual DNA molecules may be detected on the

surface of the array. Pet. 65. According to Petitioner, Figure 5 of Blumenfeld “depicts individual immobilized DNA structures 30 on the surface of the sensor.” *Id.* at 65–66 (citing Ex. 1007, Figs. 5, 8B, 9A, ¶¶ 100, 111). Petitioner further argues that “Blumenfeld incorporates by reference . . . ‘Fodor,’¹¹ . . . ‘Rava,’¹² and ‘Trulson,’¹³ which each also disclose identification of a single biomolecule, as explained by Dr. Neikirk in detail.” *Id.* at 66 (citing Ex. 1003 ¶¶ 197–201).

Patent Owner’s arguments focus on its position that Blumenfeld does not disclose an apparatus capable of identifying a single molecule, and otherwise fails to disclose all of the limitations of claim 1. Resp. 54–63. Patent Owner advances the following specific arguments.

Patent Owner argues that Blumenfeld discloses an array with spots for affixing multiple instances of a target species to detect a signal from that particular species. Resp. 54 (citing Ex. 1003 ¶¶ 194–195). Patent Owner argues that Blumenfeld does not describe an individual immobilized DNA structure, including in Figure 5. *Id.* at 54–55 (citing Ex. 2008 ¶ 228). Patent Owner argues that Blumenfeld describes DNA 30 in Figure 6 as biological material 30 made up of “DNA spots” containing multiple molecules. *Id.* at 55 (citing Ex. 1007 ¶ 102; Ex. 2008 ¶ 228). Likewise, Patent Owner argues that Blumenfeld’s Figures 8B and 9A illustrate a DNA chip that includes spots of multiple copies of a molecule, not a single molecule. *Id.* at

¹¹ Stephen P. A. Fodor et al., U.S. Patent No. 5,925,525, issued Jul. 20, 1999 (Ex. 1018, “Fodor”).

¹² Richard P. Rava et al., U.S. Patent No. 5,874,219, issued Feb. 23, 1999 (Ex. 1019, “Rava”).

¹³ Mark Trulson et al., U.S. Patent No. 5,834,758, issued Nov. 10, 1998 (Ex. 1020, “Trulson”).

56 (citing Ex. 2008 ¶ 229; Ex. 1007 ¶¶ 107, 108, 111). Patent Owner contends that Fodor, Rava, and Trulson, further describe pre-existing array technology that does not identify a single molecule. *Id.* at 56–57 (citing Ex. 2008 ¶¶ 230–233; Ex. 1003 ¶¶ 197–200).

We agree with Patent Owner that Blumenfeld does teach an apparatus that is capable of identifying a single biomolecule as required by challenged claims 1, 2, and 6. Dr. Neikirk acknowledges that “Blumenfeld discloses that it uses pre-existing array technology in which probes for target molecules are placed into ‘spots’ on the surface of the array.” Ex. 1003 ¶ 193 (citing Ex. 1007 ¶ 46). Dr. Neikirk further acknowledges that a POSA would have understood Blumenfeld teaches a “well-known technology in which each spot on the array contains probes specific for a particular molecular species, the idea being to affix multiple instances of the target species to so as to detect a signal from that particular species.” *Id.* ¶ 194.

Dr. Neikirk also asserts, however, that Figure 5 of Blumenfeld teaches identification of a single biomolecule. *Id.* We disagree. Figure 5 of Blumenfeld, set forth above, “illustrates an embodiment of an electronic light detector array detection system according to the present invention with a laser-light source.” Ex. 1007 ¶ 33. Blumenfeld further explains concerning Figure 5:

For example, the system shown in FIG. 4 is compatible with laser-biased visualization techniques as shown in FIG. 5. In other words, the image detection apparatus 10 of FIG. 4 may be modified to be in the configuration shown in FIG. 5. For instance, a laser 40 may be used to generate laser light 41 that may be reflected off of mirror 35 and thereby directed onto labeled DNA 30 hybridized and immobilized on DNA chip 14. In view of the incorporation of a fluorescent label, the hybridized DNA 30 fluoresces and emits fluorescent light upon

excitation by the laser light 41. The filter 17 selectively passes the fluorescent light but not the laser light 41. The fluorescent light is directly mapped onto the detector pixels 5 of detector 4, e.g., a CCD detector array. The laser 40 may be rastered across the biological sample, e.g., the DNA chip 14, so that only select portions of the sample are illuminated at one time. The laser 40 may be, for example, laser diodes such as green (535 nm), red (635 nm), or far red (670 nm) diodes having a power in the range of 4 mW to 10 mW.

Id. ¶ 100.

Nowhere in this discussion of Figure 5 does Blumenfeld identify DNA 30 as an individual immobilized DNA molecule. *See* Ex. 2008 ¶ 228. Figure 5, as discussed above, is offered by Blumenfeld as illustrating an electronic light detector array detection system, and Blumenfeld makes no mention that the biological samples to be detected only contain a single molecule. In fact, in the description of Figure 5 set forth above, Blumenfeld refers to the “biological sample, e.g., the DNA chip 14” and further states “that only select *portions of the sample* are illuminated at one time.” Ex. 2007 ¶ 100. Thus, it appears from this discussion of Figure 5 that “DNA 30” is a biological sample containing many DNA molecules.

Also, Blumenfeld begins the discussion of Figure 5 by stating that the system depicted in Figure 4 is compatible with the system “laser-based visualization techniques” as shown in Figure 5. Figure 4 is described as follows identifying a biological sample, not an individual molecule.

FIG. 4 is a block diagram of a system substantially similar to that of FIG. 2, wherein light from a biological sample, such as a DNA ship 14 having biological material 13 thereon, is passed through an optical filter 17 and provided to detector pixels 5 of a detector 4, e.g., a CMOS or CCD detector, where such light is sensed.

Ex. 1007 ¶ 99.

As Dr. Harris points out, Blumenfeld depicts “biological material 30” in Figure 6 with the same caret symbol (^) as shown in Figure 5 to depict “DNA 30.” *See* Ex. 1007, Figure 6. Blumenfeld refers to “biological material 30” as “DNA spots” not individual DNA molecules. *See* Ex. 1007 ¶ 102. As Dr. Harris notes, Dr. Neikirk agrees that “Blumenfeld discloses a pre-existing array technology in which probes for target molecules are placed onto ‘spots’ on the surface of an array” and that “each spot on the array contains probes specific for a particular molecular species, the idea being to affix multiple instances of the target species to [each zone] so as to detect a signal from the particular species.” Ex. 2008 ¶ 225 (citing Ex. 1003 ¶¶ 193–194). Thus, we find that Figure 5 does not support a determination that Blumenfeld teaches a device capable of identifying a single biomolecule.

Petitioner also points to Fodor, Rava, and Trulson, which Blumenfeld incorporates by reference, alleging they teach apparatuses capable of identifying a single biomolecule. Pet. 65–66. We disagree. Blumenfeld cites Fodor for technology for making chips in which the most commonly used technologies produce spots with diameters of 50-300 μm . *See* Ex. 1007 ¶ 4. Rava and Trulson are identified by Blumenfeld as providing “[f]urther descriptions of imaging fluorescently immobilized biomolecules and analysis of the images.” *Id.* ¶ 5. Petitioner has not shown persuasively that these patents were cited by Blumenfeld for any teaching relating to the amount of biological material that is identified by a biosensor.

Therefore, having considered the record as a whole, we credit Dr. Harris’ testimony that “[a] POSA would have understood Blumenfeld to be citing these patents for what it says it is citing them for: namely, prior art

methods of creating DNA chips and imaging spots on those chips.” Resp. 57 (citing Ex. 1008 ¶ 233).¹⁴

For all of these reasons set forth above, we find that Blumenfeld does not teach an apparatus for identifying a single biomolecule as required by all challenged claims. Therefore, we determine that Petitioner has failed to show by a preponderance of the evidence that claims 1, 2, and 6 are anticipated by Blumenfeld.

In the related obviousness ground, Petitioner does not offer any further argument or evidence that Blumenfeld teaches an apparatus for identifying a single biomolecule as required by the challenged claims. Therefore, we determine that Petitioner has not shown by a preponderance of the evidence that claims 1, 2, or 6 would have been obvious over Blumenfeld and the knowledge of the skilled artisan for the same reasons discussed above.

III. CONCLUSION

Based on the foregoing analysis of the arguments presented in the papers and the evidence submitted in support, we determine that Petitioner has not shown by a preponderance of the evidence that claims 1, 2, 6, 7, 10–22, 24, or 27–36 are unpatentable.

IV. ORDER

In consideration of the foregoing, it is hereby:

¹⁴ We also credit Dr. Harris’ analysis of the passages in each of Fodor, Rava, and Trulson cited by Dr. Neikirk as describing pre-existing array technology for analyzing spots containing multiple copies of biomolecules. *See* Ex. 2008 ¶¶ 230–231; *see* Ex. 2006 ¶¶ 197–199.

ORDERED that claims 1, 2, 6, 7, 10–22, 24, 27–36 of U.S. Patent 7,767,441 B2 have not been shown to be unpatentable; and

FURTHER ORDERED that because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36	102	Hassibi		1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36
1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36	103	Hassibi in view of the skill of a POSA		1, 7, 10–16, 18, 19, 21, 22, 24, 27, 30–36
17, 20	103	Hassibi, Korlach		17, 20
28, 29	103	Hassibi in view of the skill of a POSA		28, 29
21, 22, 24	103	Hassibi, Cheeseman		21, 22, 24
1, 2, 6	102	Blumenfeld		1, 2, 6
1, 2, 6	103	Blumenfeld in view of the skill of a POSA		1, 2, 6
Overall Outcome				1, 2, 6, 7, 10– 22, 24, 27–36

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